

Employee Attrition Prediction

```
In [50]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

read CSV

```
In [51]: df = pd.read_csv('hr-employee-attrition-with-null.csv')
```

Dataset statistic

```
In [52]: df.describe()
```

```
Out [52]:
```

	Unnamed: 0	Age	DailyRate	DistanceFromHome	Education	E
count	1470.000000	1176.000000	1176.000000	1176.000000	1176.000000	
mean	734.500000	37.134354	798.875850	9.37500	2.920918	
std	424.496761	9.190317	406.957684	8.23049	1.028796	
min	0.000000	18.000000	102.000000	1.00000	1.000000	
25%	367.250000	30.000000	457.750000	2.00000	2.000000	
50%	734.500000	36.000000	798.500000	7.00000	3.000000	
75%	1101.750000	43.000000	1168.250000	15.00000	4.000000	
max	1469.000000	60.000000	1499.000000	29.00000	5.000000	

8 rows × 27 columns

```
In [53]: df.head()
```

```
Out [53]:
```

	Unnamed: 0	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFr
0	0	41.0	Yes	Travel_Rarely	NaN	NaN	
1	1	NaN	No	NaN	279.0	Research & Development	
2	2	37.0	Yes	NaN	1373.0	NaN	
3	3	NaN	No	Travel_Frequently	1392.0	Research & Development	
4	4	27.0	No	Travel_Rarely	591.0	Research & Development	

5 rows × 36 columns

Feature transformation

```
In [54]: df.loc[df["Attrition"] == "no", "Attrition"] = 0.0
df.loc[df["Attrition"] == "yes", "Attrition"] = 1.0
string_categorical_col = ['Department', 'Attrition', 'BusinessTravel', 'E
                        'MaritalStatus', 'Over18', 'OverTime']

# ENCODE STRING COLUMNS TO CATEGORICAL COLUMNS
for col in string_categorical_col:
    df[col] = pd.Categorical(df[col]).codes

# HANDLE NULL NUMBERS
# Fill null numeric values with the median
numeric_cols = df.select_dtypes(include=[np.number]).columns
for col in numeric_cols:
    if df[col].isnull().any():
        df[col].fillna(df[col].median(), inplace=True)

df = df.loc[:, ~df.columns.isin(['EmployeeNumber', 'Unnamed: 0', 'Employee
```

Splitting data into train and test

```
In [55]: from sklearn.model_selection import train_test_split
```

```
In [56]: df_train, df_test = train_test_split(df, test_size=0.2, random_state=42,
```

Display histogram of each feature

```
In [57]: def display_histogram(df, col_name, cls, n_bin = 40):
    # Filter data by class
    data_class_0 = df[df['Attrition'] == 0][col_name].dropna()
    data_class_1 = df[df['Attrition'] == 1][col_name].dropna()

    # Create histogram
    plt.figure(figsize=(10, 6))
    plt.hist(data_class_0, bins=n_bin, alpha=0.5, label='No Attrition (0)')
    plt.hist(data_class_1, bins=n_bin, alpha=0.5, label='Attrition (1)',
    plt.xlabel(col_name)
    plt.ylabel('Frequency')
    plt.title(f'Histogram of {col_name} with {n_bin} bins')
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.show()
```

T4. Observe the histogram for Age, MonthlyIncome and DistanceFromHome. How many bins have zero counts? Do you think this is a good discretization? Why?

T4 Answer: With 40 bins, many bins have zero counts, especially for discrete features or features with sparse distributions. This is not ideal discretization because:

1. Zero counts lead to zero probability estimates which can cause numerical issues
2. Too many bins for continuous data can lead to overfitting
3. Bins with zero counts don't contribute useful information for classification

T5. Can we use a Gaussian to estimate this histogram? Why? What about a Gaussian Mixture Model (GMM)?

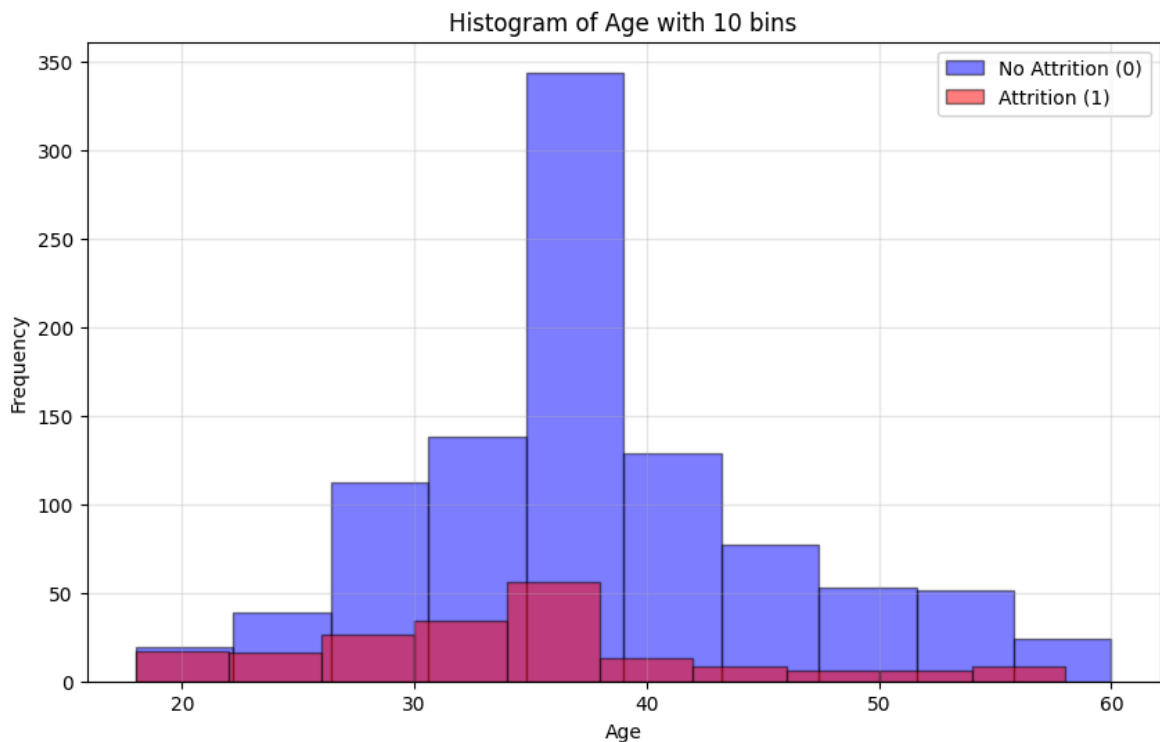
T5 Answer:

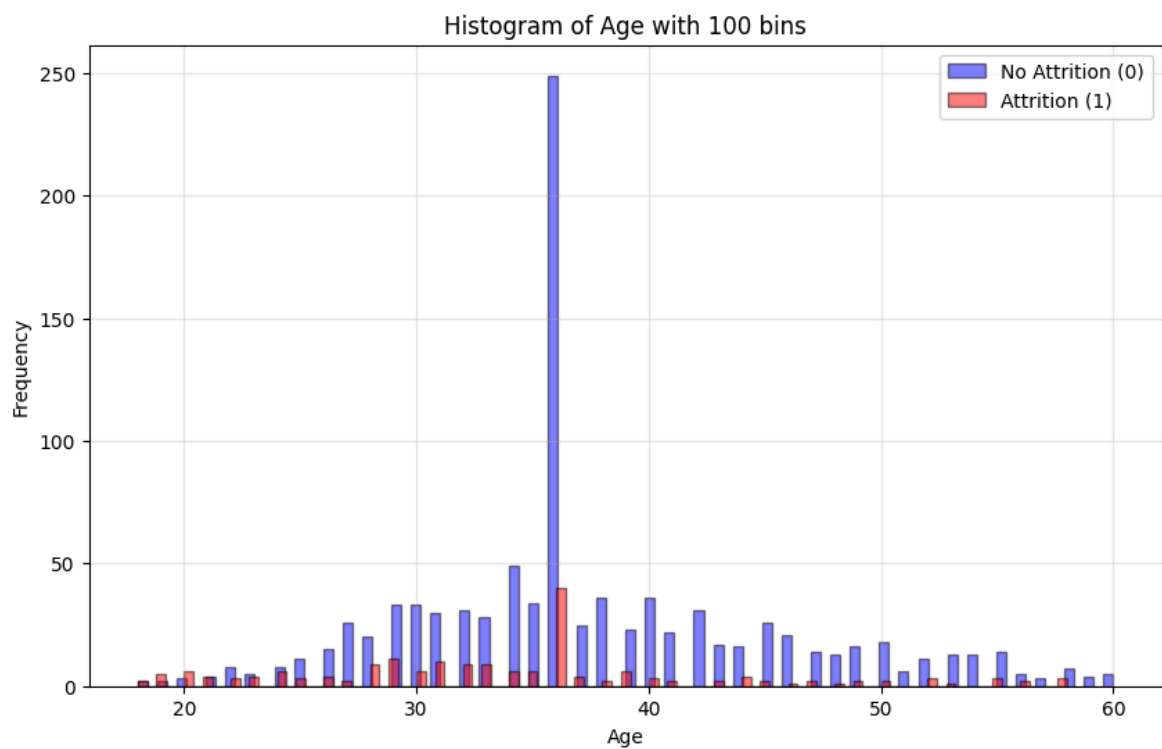
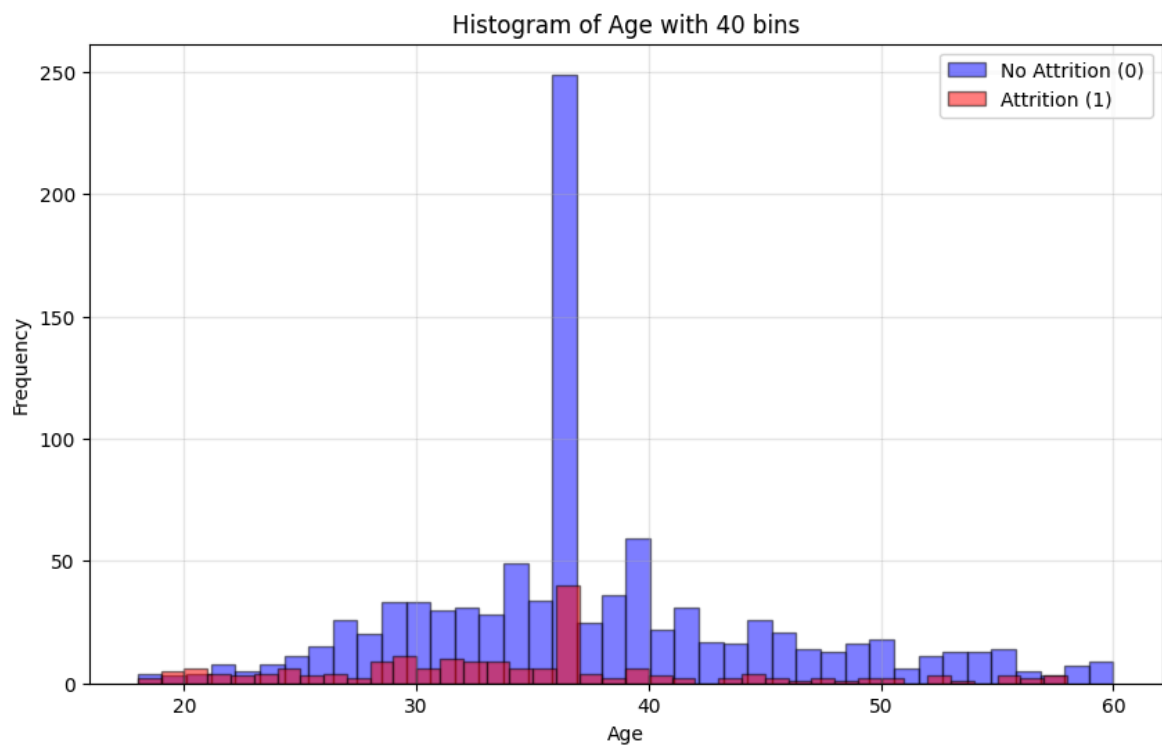
- **Single Gaussian:** Not appropriate for most of these features because the distributions are often multimodal, skewed, or have irregular shapes that cannot be captured by a single bell curve.
- **Gaussian Mixture Model (GMM):** Yes, GMM can be used as it can model complex distributions by combining multiple Gaussians. This allows it to capture multimodal distributions and more complex patterns in the data.

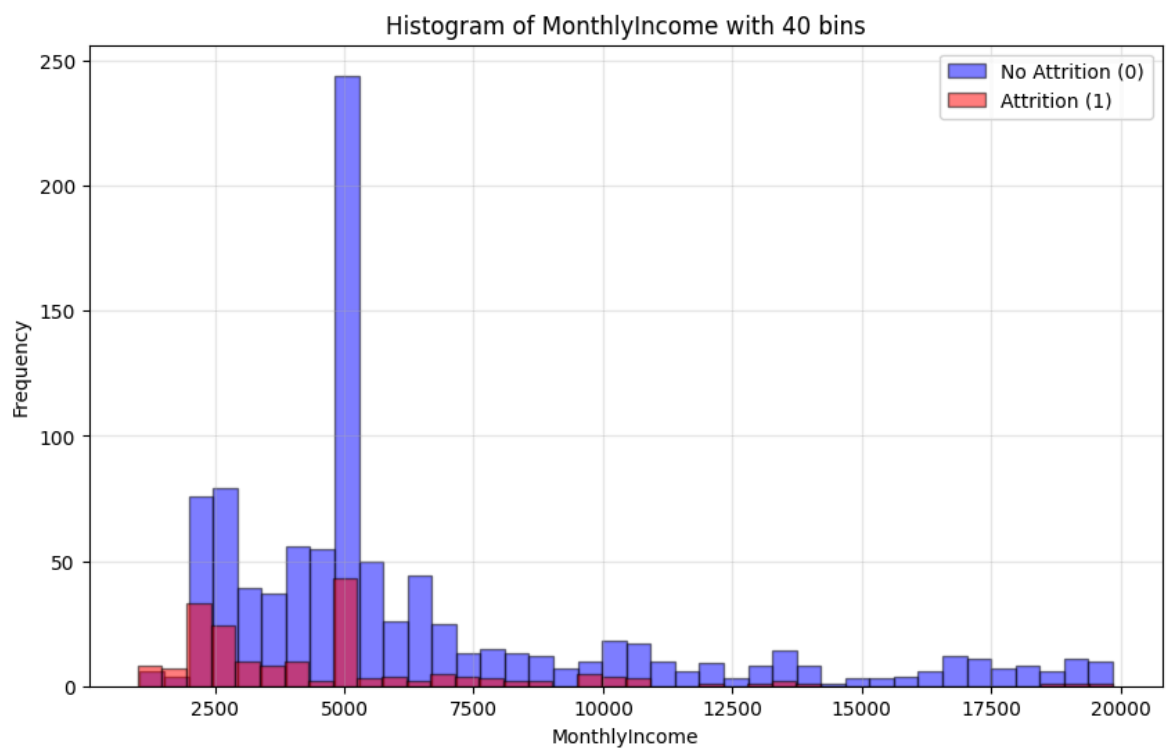
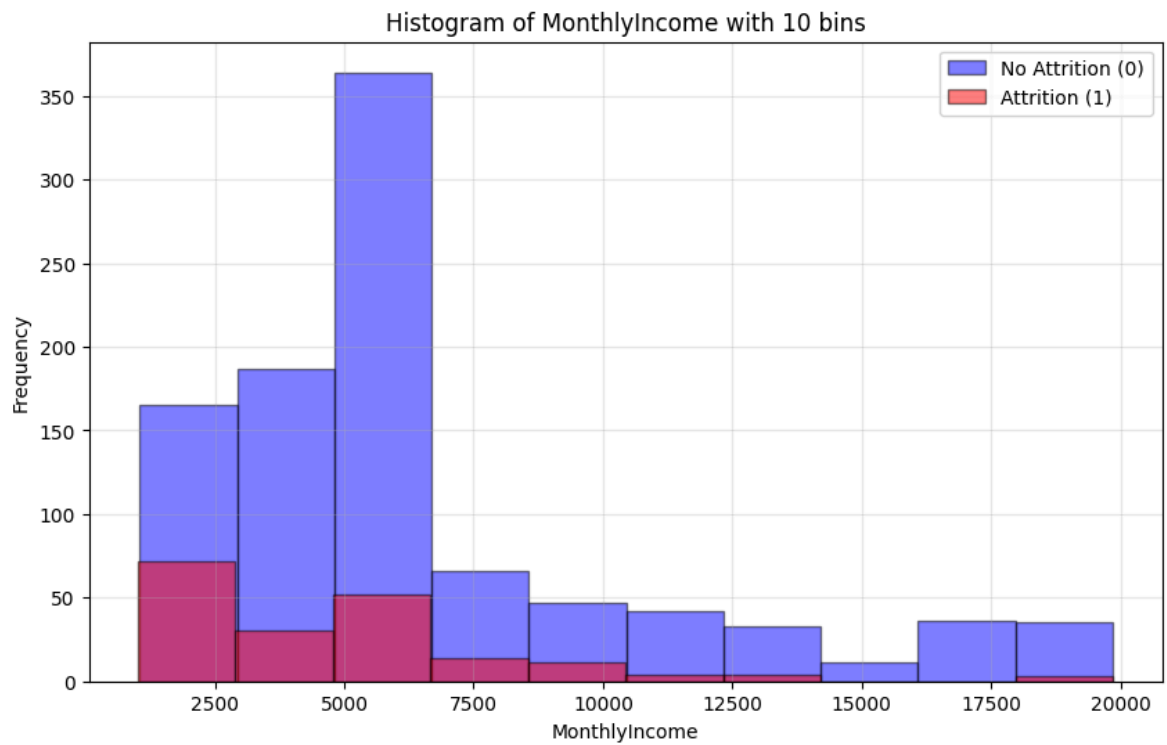
T6. Now plot the histogram according to the method described above (with 10, 40, and 100 bins) and show 3 plots each for Age, MonthlyIncome, and DistanceFromHome. Which bin size is most sensible for each features? Why?

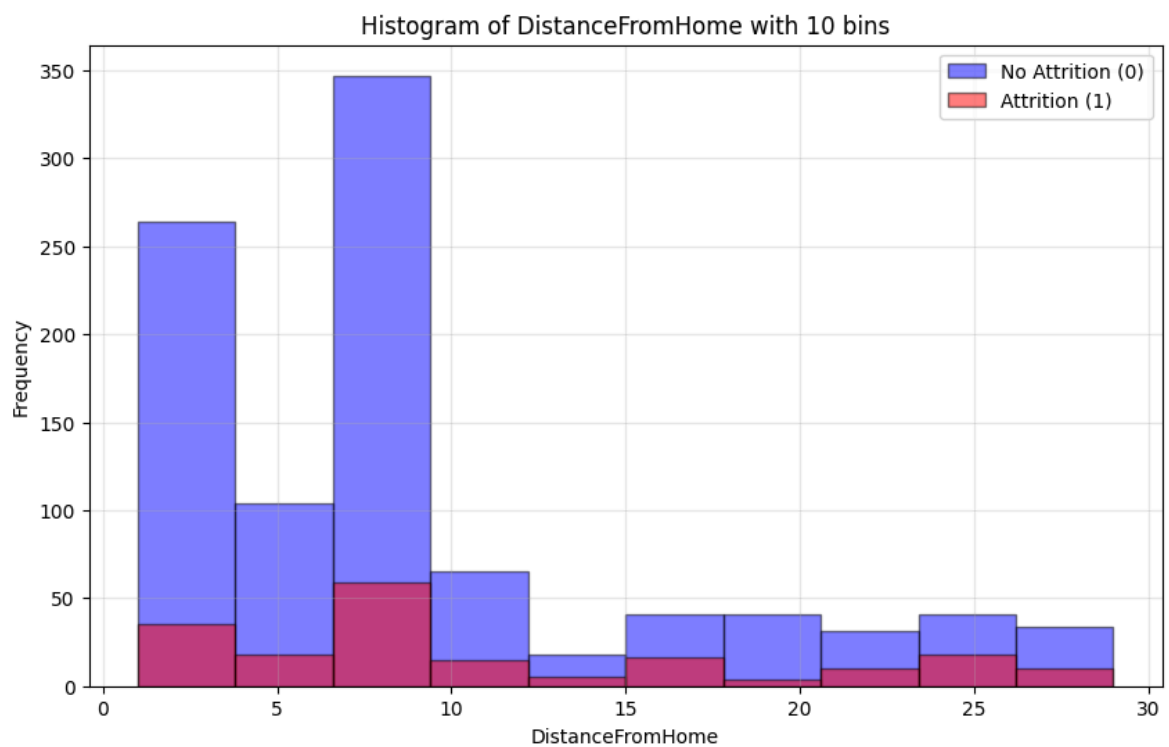
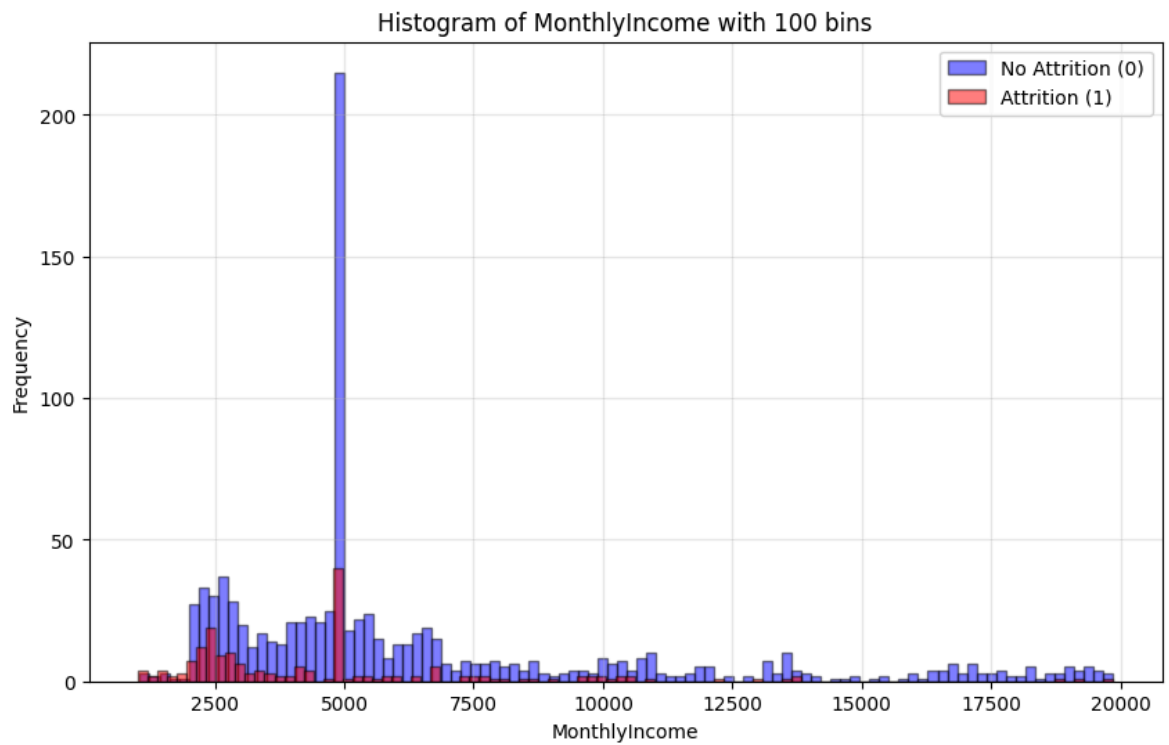
```
In [58]: # T6: Plot histograms with different bin sizes
features_to_plot = ['Age', 'MonthlyIncome', 'DistanceFromHome']
bin_sizes = [10, 40, 100]

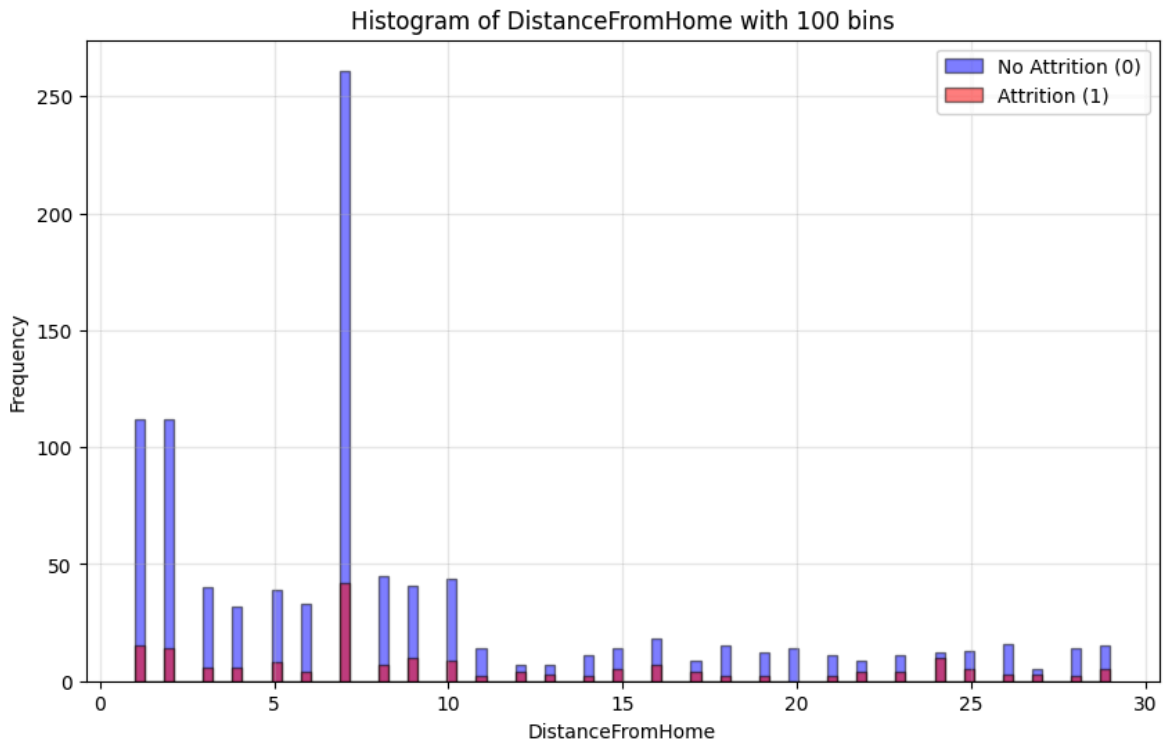
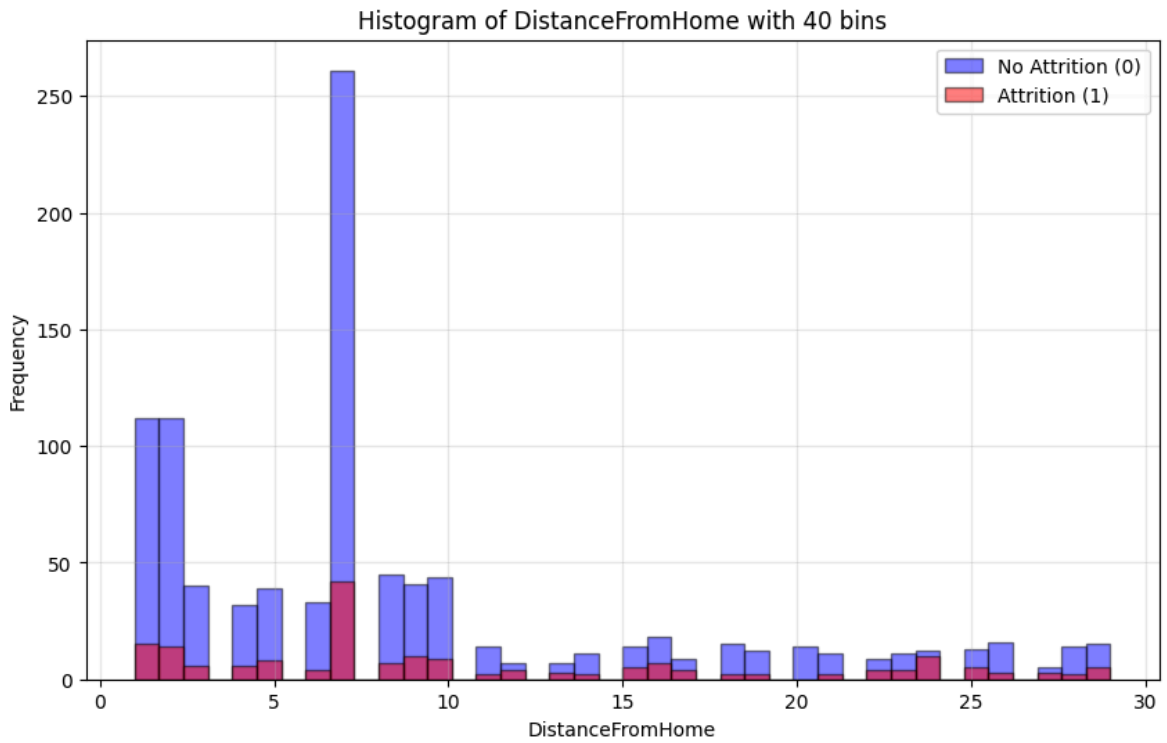
for feature in features_to_plot:
    if feature in df_train.columns:
        for n_bins in bin_sizes:
            display_histogram(df_train, feature, 'Attrition', n_bin=n_bin
```











T6 Answer:

- **Age:** 10-40 bins is most sensible. Age has a reasonable range (~18-60) and a moderate number of bins captures the distribution without creating too many empty bins.
- **MonthlyIncome:** 10-40 bins is appropriate. Income varies widely but too many bins create sparsity.
- **DistanceFromHome:** 10 bins is sufficient. This feature has a smaller range and fewer distinct values, so fewer bins prevent overfitting.

The key is balancing between capturing the distribution shape and avoiding sparse bins with zero counts.

T7. For the rest of the features, which one should be discretized in order to be modeled by histograms? What are the criteria for choosing whether we should discretize a feature or not? Answer this and discretize those features into 10 bins each. In other words, figure out the bin edge for each feature, then use digitize() to convert the features to discrete values

T7 Answer:

Features that should be discretized:

- Continuous features with many distinct values should be discretized (e.g., Age, MonthlyIncome, DistanceFromHome, HourlyRate, DailyRate, MonthlyRate, etc.)

Criteria for discretization:

1. The feature has many distinct continuous or integer values
2. The feature is not already categorical
3. Discretization can help capture non-linear relationships
4. The number of distinct values is large relative to sample size

Features that should NOT be discretized:

- Already categorical features (e.g., JobRole, Department, MaritalStatus)
- Binary features (e.g., Gender, OverTime)
- Features with very few distinct values already

T8. What kind of distribution should we use to model histograms? (Answer a distribution name) What is the MLE for the likelihood distribution? (Describe how to do the MLE). Plot the likelihood distributions of MonthlyIncome, JobRole, HourlyRate, and MaritalStatus for different Attrition values.

T8 Answer:

Distribution type: Categorical/Multinomial distribution (or Discrete distribution)

MLE for likelihood distribution: For each bin i and class c , the MLE is:

$$P(x_i|c) = \frac{\text{count}(x_i, c)}{N_c}$$

where:

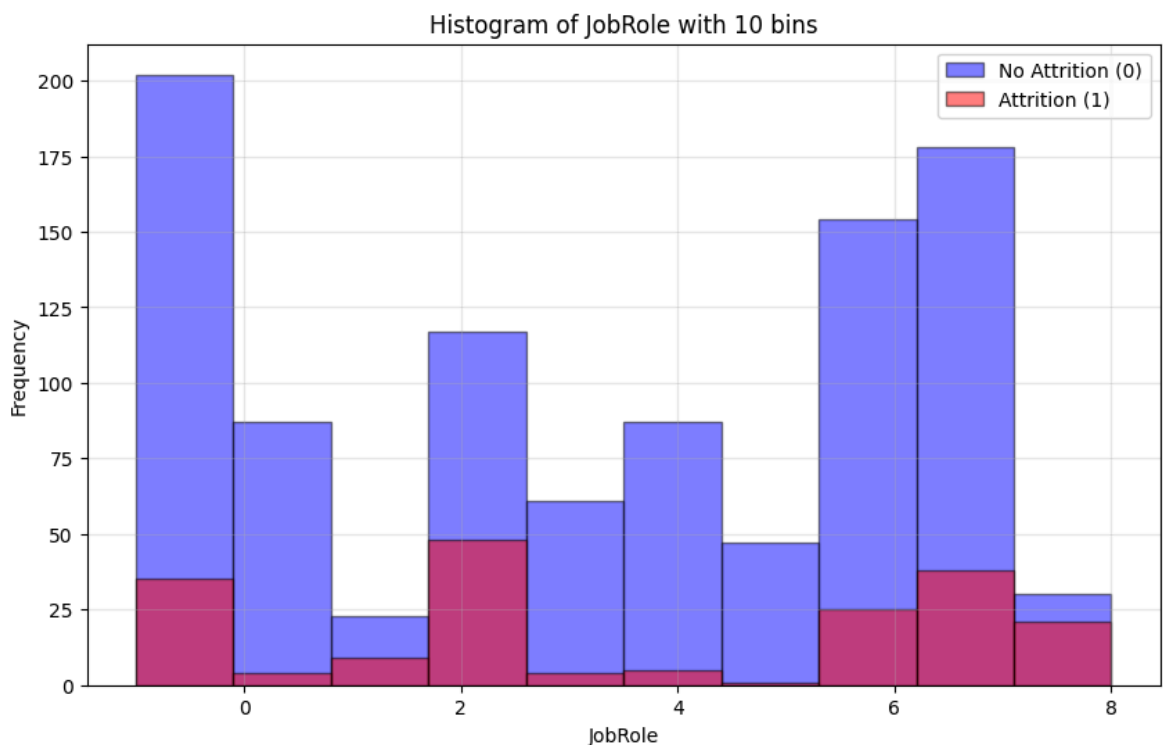
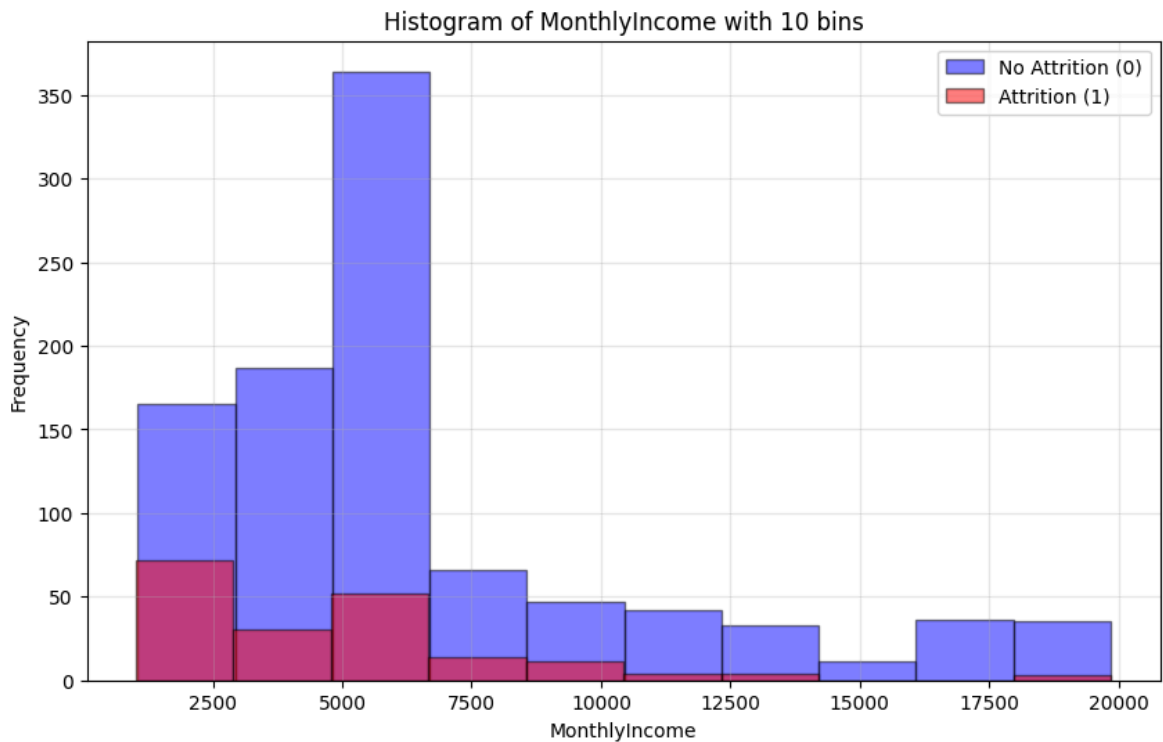
- $\text{count}(x_i, c)$ is the number of samples in bin i for class c
- N_c is the total number of samples in class c

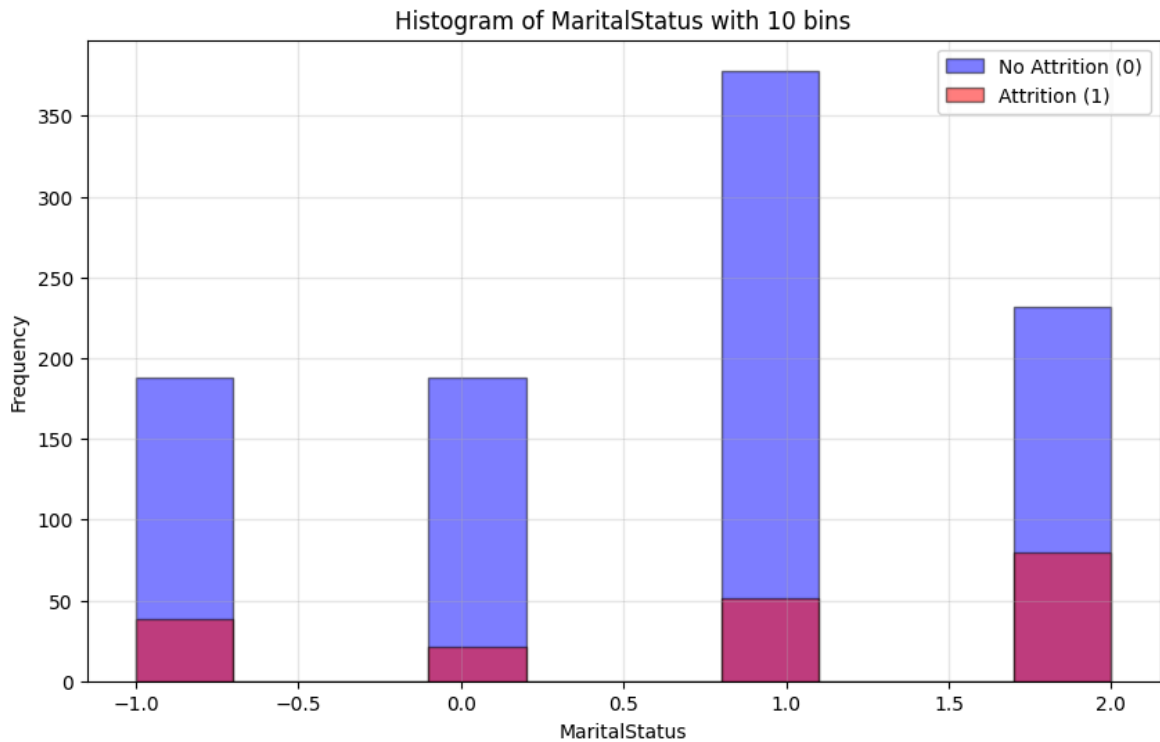
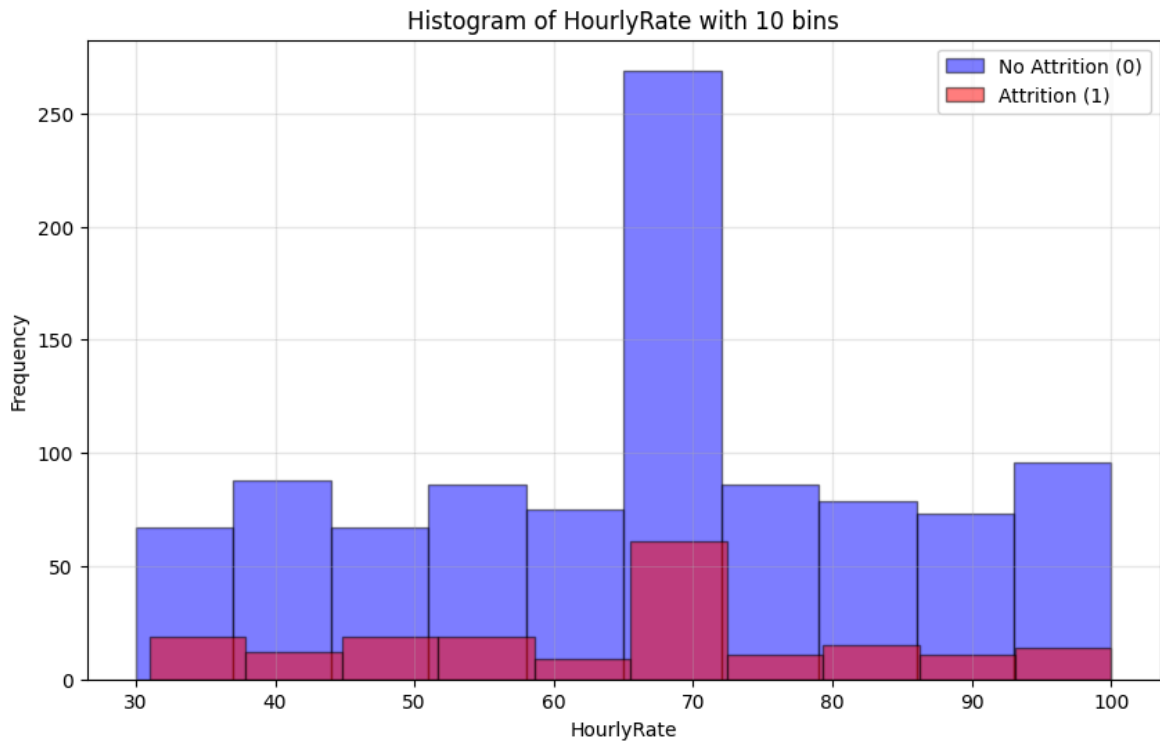
To avoid zero probabilities, we use Laplace smoothing (add-one smoothing):

$$P(x_i|c) = \frac{\text{count}(x_i, c) + 1}{N_c + K}$$

where K is the number of bins.

```
In [59]: # T8: Plot likelihood distributions
features_to_plot_t8 = ['MonthlyIncome', 'JobRole', 'HourlyRate', 'Marital
for feature in features_to_plot_t8:
    if feature in df_train.columns:
        display_histogram(df_train, feature, 'Attrition', n_bin=10)
```





T10. If we use the current Naive Bayes with our current Maximum Likelihood Estimates, we will find that some $P(x_i | \text{attrition})$ will be zero and will result in the entire product term to be zero. Propose a method to fix this problem.

T10 Answer:

Solution: Laplace Smoothing (Add-one smoothing)

When a feature value appears in the test set but never appeared with a particular class in the training set, the probability $P(x_i | \text{class}) = 0$, making the entire product zero.

Fix: Add a small constant (typically 1) to all counts:

$$P(x_i|c) = \frac{\text{count}(x_i, c) + \alpha}{N_c + \alpha \times K}$$

where:

- α is the smoothing parameter (usually 1)
- K is the number of possible values/bins
- This ensures no probability is exactly zero

This method is implemented in the SimpleBayesClassifier by adding 1 to all bin counts during the fit_params method.

T11. Implement your Naive Bayes classifier. Use the learned distributions to classify the test set. Don't forget to allow your classifier to handle missing values in the test set. Report the overall Accuracy. Then, report the Precision, Recall, and F score for detecting attrition. See Lecture 1 for the definitions of each metric.

```
In [60]: from SimpleBayesClassifier import SimpleBayesClassifier

In [61]: data_train = df_train.to_numpy()
data_test = df_test.to_numpy()

In [62]: x_train = data_train[:, :-1] # All columns except last (Attrition)
y_train = data_train[:, -1] # Last column (Attrition)

x_test = data_test[:, :-1]
y_test = data_test[:, -1]

In [63]: model = SimpleBayesClassifier(n_pos=int(np.sum(y_train == 1)), n_neg=int(

In [64]: def check_prior():
        """
        This function designed to test the implementation of the prior probab
        Specifically, it checks if the classifier correctly computes the prio
        negative and positive classes based on given input counts.
        """

        # prior_neg = 5/(5 + 5) = 0.5 and # prior_pos = 5/(5 + 5) = 0.5
        assert (SimpleBayesClassifier(5, 5).prior_pos, SimpleBayesClassifier(

        assert (SimpleBayesClassifier(3, 5).prior_pos, SimpleBayesClassifier(
        assert (SimpleBayesClassifier(0, 1).prior_pos, SimpleBayesClassifier(
        assert (SimpleBayesClassifier(1, 0).prior_pos, SimpleBayesClassifier(

        check_prior()

In [65]: model.fit_params(x_train, y_train)
```

```

Out[65]: ([array([17, 11, 21, 19, 58, 23, 10, 9, 3, 5]),
          array([18. , 22.2, 26.4, 30.6, 34.8, 39. , 43.2, 47.4, 51.6, 55.8, 6
0. ])),
          (array([108, 1, 1, 1, 1, 1, 1, 1, 1, 60]),
          array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. ])),
          (array([29, 1, 1, 16, 1, 1, 29, 1, 1, 96]),
          array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
          (array([19, 13, 20, 19, 44, 12, 10, 15, 13, 11]),
          array([ 104. , 243.1, 382.2, 521.3, 660.4, 799.5, 938.6, 1077.
7,
1216.8, 1355.9, 1495. ])),
          (array([42, 1, 1, 9, 1, 1, 79, 1, 1, 40]),
          array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
          (array([47, 16, 52, 13, 5, 9, 6, 9, 12, 7]),
          array([ 1. , 3.8, 6.6, 9.4, 12.2, 15. , 17.8, 20.6, 23.4, 26.2, 2
9. ])),
          (array([19, 1, 26, 1, 92, 1, 1, 27, 1, 7]),
          array([1. , 1.4, 1.8, 2.2, 2.6, 3. , 3.4, 3.8, 4.2, 4.6, 5. ])),
          (array([37, 4, 1, 54, 21, 1, 39, 1, 6, 12]),
          array([-1. , -0.4, 0.2, 0.8, 1.4, 2. , 2.6, 3.2, 3.8, 4.4,
5. ])),
          (array([30, 1, 1, 26, 1, 1, 72, 1, 1, 42]),
          array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
          (array([33, 1, 1, 1, 51, 1, 1, 1, 1, 85]),
          array([-1.00000000e+00, -8.00000000e-01, -6.00000000e-01, -4.00000000
e-01,
-2.00000000e-01, 5.00000041e-11, 2.00000000e-01, 4.00000000
e-01,
6.00000000e-01, 8.00000000e-01, 1.00000000e+00])),
          (array([16, 10, 14, 13, 13, 48, 18, 15, 15, 14]),
          array([ 30., 37., 44., 51., 58., 65., 72., 79., 86., 93., 10
0. ])),
          (array([ 13, 1, 1, 35, 1, 1, 110, 1, 1, 12]),
          array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
          (array([78, 1, 59, 1, 19, 1, 1, 10, 1, 5]),
          array([1. , 1.4, 1.8, 2.2, 2.6, 3. , 3.4, 3.8, 4.2, 4.6, 5. ])),
          (array([41, 10, 6, 34, 8, 8, 6, 23, 21, 19]),
          array([-1. , -0.1, 0.8, 1.7, 2.6, 3.5, 4.4, 5.3, 6.2, 7.1,
8. ])),
          (array([19, 1, 1, 26, 1, 1, 78, 1, 1, 47]),
          array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
          (array([32, 1, 1, 27, 1, 1, 59, 1, 1, 52]),
          array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
          (array([69, 22, 44, 12, 5, 6, 5, 3, 4, 6]),
          array([ 1009. , 2841.2, 4673.4, 6505.6, 8337.8, 10170. , 12002.2,
13834.4, 15666.6, 17498.8, 19331. ])),
          (array([14, 13, 15, 12, 52, 20, 13, 13, 15, 9]),
          array([ 2097. , 4569.3, 7041.6, 9513.9, 11986.2, 14458.5, 16930.8,
19403.1, 21875.4, 24347.7, 26820. ])),
          (array([10, 51, 42, 19, 19, 7, 8, 8, 5, 7]),
          array([0. , 0.9, 1.8, 2.7, 3.6, 4.5, 5.4, 6.3, 7.2, 8.1, 9. ])),
          (array([31, 1, 1, 1, 97, 1, 1, 1, 1, 41]),
          array([-1.00000000e+00, -8.00000000e-01, -6.00000000e-01, -4.00000000
e-01,
-2.00000000e-01, 5.00000041e-11, 2.00000000e-01, 4.00000000
e-01,
6.00000000e-01, 8.00000000e-01, 1.00000000e+00])),

```

```

(array([35, 18, 60, 8, 16, 10, 5, 12, 7, 5]),
 array([11. , 12.4, 13.8, 15.2, 16.6, 18. , 19.4, 20.8, 22.2, 23.6, 2
5. ])),
(array([139, 1, 1, 1, 1, 1, 1, 1, 1, 29]),
 array([3. , 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4. ])),
(array([23, 1, 1, 29, 1, 1, 80, 1, 1, 38]),
 array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([68, 1, 1, 81, 1, 1, 13, 1, 1, 8]),
 array([0. , 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, 2.4, 2.7, 3. ])),
(array([47, 17, 21, 55, 10, 9, 7, 4, 2, 4]),
 array([ 0. , 3.2, 6.4, 9.6, 12.8, 16. , 19.2, 22.4, 25.6, 28.8, 3
2. ])),
(array([ 7, 7, 1, 57, 65, 1, 16, 1, 12, 9]),
 array([0. , 0.6, 1.2, 1.8, 2.4, 3. , 3.6, 4.2, 4.8, 5.4, 6. ])),
(array([ 9, 1, 1, 29, 1, 1, 110, 1, 1, 22]),
 array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([100, 4, 4, 50, 3, 5, 4, 3, 1, 2]),
 array([ 0. , 1.3, 2.6, 3.9, 5.2, 6.5, 7.8, 9.1, 10.4, 11.7, 1
3. ])),
(array([100, 13, 41, 3, 2, 7, 5, 2, 1, 2]),
 array([ 0. , 1.2, 2.4, 3.6, 4.8, 6. , 7.2, 8.4, 9.6, 10.8, 1
2. ])),
(array([160, 4, 2, 1, 1, 3, 1, 1, 2, 1]),
 array([ 0. , 1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 7.7, 8.8, 9.9, 1
1. ]))),
[(array([ 5, 4, 3, 7, 18, 6, 5, 2, 1, 2]),
 array([18. , 22.2, 26.4, 30.6, 34.8, 39. , 43.2, 47.4, 51.6, 55.8, 6
0. ])),
(array([37, 1, 1, 1, 1, 1, 1, 1, 1, 8]),
 array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. ])),
(array([ 9, 1, 1, 3, 1, 1, 9, 1, 1, 26]),
 array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
(array([3, 6, 4, 4, 7, 5, 3, 7, 8, 6]),
 array([ 104. , 243.1, 382.2, 521.3, 660.4, 799.5, 938.6, 1077.
7,
1216.8, 1355.9, 1495. ])),
(array([13, 1, 1, 1, 1, 1, 23, 1, 1, 10]),
 array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
(array([11, 1, 19, 6, 2, 1, 2, 1, 7, 3]),
 array([ 1. , 3.8, 6.6, 9.4, 12.2, 15. , 17.8, 20.6, 23.4, 26.2, 2
9. ])),
(array([ 7, 1, 9, 1, 16, 1, 1, 13, 1, 3]),
 array([1. , 1.4, 1.8, 2.2, 2.6, 3. , 3.4, 3.8, 4.2, 4.6, 5. ])),
(array([ 9, 1, 1, 16, 4, 1, 12, 1, 5, 3]),
 array([-1. , -0.4, 0.2, 0.8, 1.4, 2. , 2.6, 3.2, 3.8, 4.4,
5. ])),
(array([11, 1, 1, 8, 1, 1, 18, 1, 1, 10]),
 array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([ 9, 1, 1, 1, 15, 1, 1, 1, 1, 22]),
 array([-1.00000000e+00, -8.00000000e-01, -6.00000000e-01, -4.00000000
e-01,
-2.00000000e-01, 5.00000041e-11, 2.00000000e-01, 4.00000000
e-01,
6.00000000e-01, 8.00000000e-01, 1.00000000e+00])),
(array([ 4, 6, 8, 5, 3, 14, 2, 5, 2, 4]),
 array([ 30., 37., 44., 51., 58., 65., 72., 79., 86., 93., 10
0.])),
(array([ 4, 1, 1, 7, 1, 1, 33, 1, 1, 3]),

```

```

array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([15, 1, 19, 1, 6, 1, 1, 3, 1, 5])),
array([1. , 1.4, 1.8, 2.2, 2.6, 3. , 3.4, 3.8, 4.2, 4.6, 5. ])),
(array([ 8, 3, 2, 8, 4, 3, 4, 8, 12, 1])),
array([-1. , -0.1, 0.8, 1.7, 2.6, 3.5, 4.4, 5.3, 6.2, 7.1,
8. ])),
(array([ 8, 1, 1, 4, 1, 1, 19, 1, 1, 16])),
array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([10, 1, 1, 4, 1, 1, 16, 1, 1, 17])),
array([-1. , -0.7, -0.4, -0.1, 0.2, 0.5, 0.8, 1.1, 1.4, 1.7,
2. ])),
(array([14, 4, 15, 4, 2, 5, 2, 1, 2, 4])),
array([ 1009. , 2841.2, 4673.4, 6505.6, 8337.8, 10170. , 12002.2,
13834.4, 15666.6, 17498.8, 19331. ])),
(array([ 4, 6, 4, 3, 16, 5, 5, 4, 3, 3])),
array([ 2097. , 4569.3, 7041.6, 9513.9, 11986.2, 14458.5, 16930.8,
19403.1, 21875.4, 24347.7, 26820. ])),
(array([ 5, 15, 9, 6, 4, 3, 4, 3, 3, 1])),
array([0. , 0.9, 1.8, 2.7, 3.6, 4.5, 5.4, 6.3, 7.2, 8.1, 9. ])),
(array([11, 1, 1, 1, 27, 1, 1, 1, 1, 8])),
array([-1.00000000e+00, -8.00000000e-01, -6.00000000e-01, -4.00000000
e-01,
-2.00000000e-01, 5.00000041e-11, 2.00000000e-01, 4.00000000
e-01,
6.00000000e-01, 8.00000000e-01, 1.00000000e+00])),
(array([14, 7, 12, 3, 6, 2, 1, 5, 1, 2])),
array([11. , 12.4, 13.8, 15.2, 16.6, 18. , 19.4, 20.8, 22.2, 23.6, 2
5. ])),
(array([41, 1, 1, 1, 1, 1, 1, 1, 1, 1, 4])),
array([3. , 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4. ])),
(array([ 6, 1, 1, 5, 1, 1, 22, 1, 1, 14])),
array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([24, 1, 1, 20, 1, 1, 2, 1, 1, 1])),
array([0. , 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, 2.4, 2.7, 3. ])),
(array([ 6, 7, 8, 15, 2, 4, 3, 2, 4, 2])),
array([ 0. , 3.2, 6.4, 9.6, 12.8, 16. , 19.2, 22.4, 25.6, 28.8, 3
2. ])),
(array([ 3, 2, 1, 14, 20, 1, 2, 1, 5, 4])),
array([0. , 0.6, 1.2, 1.8, 2.4, 3. , 3.6, 4.2, 4.8, 5.4, 6. ])),
(array([ 2, 1, 1, 13, 1, 1, 28, 1, 1, 4])),
array([1. , 1.3, 1.6, 1.9, 2.2, 2.5, 2.8, 3.1, 3.4, 3.7, 4. ])),
(array([ 8, 6, 3, 20, 4, 2, 4, 3, 2, 1])),
array([ 0. , 1.3, 2.6, 3.9, 5.2, 6.5, 7.8, 9.1, 10.4, 11.7, 1
3. ])),
(array([10, 13, 12, 4, 2, 6, 2, 1, 2, 1])),
array([ 0. , 1.2, 2.4, 3.6, 4.8, 6. , 7.2, 8.4, 9.6, 10.8, 1
2. ])),
(array([33, 7, 1, 2, 1, 1, 3, 2, 1, 2])),
array([ 0. , 1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 7.7, 8.8, 9.9, 1
1. ])))]

```

In [66]: `def check_fit_params():`

"""

This function is designed to test the fit_params method of a SimpleBa
This method is presumably responsible for computing parameters for a
based on the provided training data. The parameters in this context i
"""

T = SimpleBayesClassifier(2, 2)

```

X_TRAIN_CASE_1 = np.array([
    [0, 1, 2, 3],
    [1, 2, 3, 4],
    [2, 3, 4, 5],
    [3, 4, 5, 6]
])
Y_TRAIN_CASE_1 = np.array([0, 1, 0, 1])
STAY_PARAMS_1, LEAVE_PARAMS_1 = T.fit_params(X_TRAIN_CASE_1, Y_TRAIN_

print("STAY PARAMETERS")
for f_idx in range(len(STAY_PARAMS_1)):
    print(f"Feature : {f_idx}")
    print(f"BINS : {STAY_PARAMS_1[f_idx][0]}")
    print(f"EDGES : {STAY_PARAMS_1[f_idx][1]}")
print("")
print("LEAVE PARAMETERS")
for f_idx in range(len(STAY_PARAMS_1)):
    print(f"Feature : {f_idx}")
    print(f"BINS : {LEAVE_PARAMS_1[f_idx][0]}")
    print(f"EDGES : {LEAVE_PARAMS_1[f_idx][1]}")

check_fit_params()

```

STAY PARAMETERS

Feature : 0

BINS : [2 1 1 1 1 1 2 1 1 1]

EDGES : [0. 0.3 0.6 0.9 1.2 1.5 1.8 2.1 2.4 2.7 3.]

Feature : 1

BINS : [2 1 1 1 1 1 2 1 1 1]

EDGES : [1. 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.]

Feature : 2

BINS : [2 1 1 1 1 1 2 1 1 1]

EDGES : [2. 2.3 2.6 2.9 3.2 3.5 3.8 4.1 4.4 4.7 5.]

Feature : 3

BINS : [2 1 1 1 1 1 2 1 1 1]

EDGES : [3. 3.3 3.6 3.9 4.2 4.5 4.8 5.1 5.4 5.7 6.]

LEAVE PARAMETERS

Feature : 0

BINS : [1 1 1 2 1 1 1 1 1 2]

EDGES : [0. 0.3 0.6 0.9 1.2 1.5 1.8 2.1 2.4 2.7 3.]

Feature : 1

BINS : [1 1 1 2 1 1 1 1 1 2]

EDGES : [1. 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.]

Feature : 2

BINS : [1 1 1 2 1 1 1 1 1 2]

EDGES : [2. 2.3 2.6 2.9 3.2 3.5 3.8 4.1 4.4 4.7 5.]

Feature : 3

BINS : [1 1 1 2 1 1 1 1 1 2]

EDGES : [3. 3.3 3.6 3.9 4.2 4.5 4.8 5.1 5.4 5.7 6.]

In [67]: `y_pred = model.predict(x = x_test)`

In [68]: `def evaluate(y_true, y_pred, show_result = True):`
`# Calculate metrics`
`TP = np.sum((y_true == 1) & (y_pred == 1))`
`TN = np.sum((y_true == 0) & (y_pred == 0))`
`FP = np.sum((y_true == 0) & (y_pred == 1))`
`FN = np.sum((y_true == 1) & (y_pred == 0))`

```

accuracy = (TP + TN) / (TP + TN + FP + FN) if (TP + TN + FP + FN) > 0
precision = TP / (TP + FP) if (TP + FP) > 0 else 0
recall = TP / (TP + FN) if (TP + FN) > 0 else 0
F1 = 2 * (precision * recall) / (precision + recall) if (precision + recall) > 0 else 0
fpr = FP / (FP + TN) if (FP + TN) > 0 else 0

if show_result:
    print(f"Accuracy: {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall: {recall:.4f}")
    print(f"F1 Score: {F1:.4f}")
    print(f"FPR: {fpr:.4f}")

return accuracy, precision, recall, F1, fpr

```

In [69]: `evaluate(y_test, y_pred)`

```

Accuracy: 0.8226
Precision: 0.7500
Recall: 0.2308
F1 Score: 0.3529
FPR: 0.0204

```

Out [69]: `(np.float64(0.8225806451612904),
np.float64(0.75),
np.float64(0.23076923076923078),
np.float64(0.3529411764705882),
np.float64(0.02040816326530612))`

T9. What is the prior distribution of the two classes?

```

In [70]: # T9: Calculate prior distribution
n_total = len(y_train)
n_attrition = np.sum(y_train == 1)
n_no_attrition = np.sum(y_train == 0)

prior_attrition = n_attrition / n_total
prior_no_attrition = n_no_attrition / n_total

print(f"Prior P(Attrition=1) = {prior_attrition:.4f}")
print(f"Prior P(Attrition=0) = {prior_no_attrition:.4f}")

```

```

Prior P(Attrition=1) = 0.0366
Prior P(Attrition=0) = 0.1412

```

T12. Use the learned distributions to classify the test set. Report the results using the same metric as the previous question.

In [71]: `model.fit_gaussian_params(x_train, y_train)`

```

Out[71]: ([ (np.float64(35.204819277108435), np.float64(8.695323540464916)),
            (np.float64(0.35542168674698793), np.float64(0.4786409001923194)),
            (np.float64(1.144578313253012), np.float64(1.1473650286318104)),
            (np.float64(745.855421686747), np.float64(366.32734004025247)),
            (np.float64(0.6927710843373494), np.float64(1.084755608757175)),
            (np.float64(9.373493975903614), np.float64(7.907920146608996)),
            (np.float64(2.86144578313253), np.float64(0.9307765025305168)),
            (np.float64(1.4819277108433735), np.float64(1.738032718251885)),
            (np.float64(2.746987951807229), np.float64(1.0160563201112296)),
            (np.float64(0.3132530120481928), np.float64(0.7750275680036333)),
            (np.float64(66.355421686747), np.float64(17.815322719989133)),
            (np.float64(2.716867469879518), np.float64(0.6928758433103265)),
            (np.float64(1.8253012048192772), np.float64(0.9876762765946878)),
            (np.float64(3.13855421686747), np.float64(3.2615072029680294)),
            (np.float64(2.9096385542168677), np.float64(0.9237318833127457)),
            (np.float64(0.7771084337349398), np.float64(1.077572606918849)),
            (np.float64(4979.183734939759), np.float64(4030.2182260611157)),
            (np.float64(14095.277108433735), np.float64(5900.313825252343)),
            (np.float64(2.8012048192771086), np.float64(2.2607704186316457)),
            (np.float64(0.060240963855421686), np.float64(0.6465738730119874)),
            (np.float64(15.331325301204819), np.float64(3.5658834458799706)),
            (np.float64(3.1686746987951806), np.float64(0.3744643438053025)),
            (np.float64(2.789156626506024), np.float64(0.9366065443085648)),
            (np.float64(0.7530120481927711), np.float64(0.7639209808056603)),
            (np.float64(8.614457831325302), np.float64(6.688114480813167)),
            (np.float64(2.8493975903614457), np.float64(1.2544737766807785)),
            (np.float64(2.86144578313253), np.float64(0.6846565096580605)),
            (np.float64(2.5963855421686746), np.float64(2.531676228520549)),
            (np.float64(1.5662650602409638), np.float64(2.263874337078014)),
            (np.float64(0.46987951807228917), np.float64(1.05667527466413))]
          [(np.float64(35.13953488372093), np.float64(7.783648320615192)),
            (np.float64(0.16279069767441862), np.float64(0.36917460154389636)),
            (np.float64(1.1627906976744187), np.float64(1.1599966431013884)),
            (np.float64(883.8837209302326), np.float64(389.63636650996136)),
            (np.float64(0.6511627906976745), np.float64(1.0972210336823673)),
            (np.float64(10.232558139534884), np.float64(8.24575213911036)),
            (np.float64(2.9069767441860463), np.float64(1.0957413013385813)),
            (np.float64(1.6744186046511629), np.float64(1.721558653322553)),
            (np.float64(2.5813953488372094), np.float64(1.0616476474896117)),
            (np.float64(0.3023255813953488), np.float64(0.7635560539247443)),
            (np.float64(61.48837209302326), np.float64(17.2954787653309)),
            (np.float64(2.7674418604651163), np.float64(0.6411185466088475)),
            (np.float64(2.1627906976744184), np.float64(1.1994229728583838)),
            (np.float64(3.697674418604651), np.float64(2.9455120273560387)),
            (np.float64(2.953488372093023), np.float64(1.0332492972801424)),
            (np.float64(0.8837209302325582), np.float64(1.124966197437007)),
            (np.float64(6229.790697674419), np.float64(4666.9164041832155)),
            (np.float64(13847.093023255815), np.float64(6033.466598950309)),
            (np.float64(2.6744186046511627), np.float64(2.238364542421575)),
            (np.float64(-0.06976744186046512), np.float64(0.6248850624439257)),
            (np.float64(14.790697674418604), np.float64(3.427373660826147)),
            (np.float64(3.0697674418604652), np.float64(0.25475467790937956)),
            (np.float64(2.9767441860465116), np.float64(0.9273210249942737)),
            (np.float64(0.4883720930232558), np.float64(0.5444046470102151)),
            (np.float64(11.162790697674419), np.float64(7.35773673668915)),
            (np.float64(2.9302325581395348), np.float64(1.3536402056609533)),
            (np.float64(2.744186046511628), np.float64(0.6135305096871125)),
            (np.float64(4.558139534883721), np.float64(2.581709597465659)),
            (np.float64(2.953488372093023), np.float64(2.4299821836086553)),
            (np.float64(1.4651162790697674), np.float64(2.385506988279502))]])

```

```
In [72]: def check_fit_gaussian_params():

        """
        This function is designed to test the fit_gaussian_params method of a
        This method is presumably responsible for computing parameters for a
        based on the provided training data. The parameters in this context i
        """

        T = SimpleBayesClassifier(2, 2)
        X_TRAIN_CASE_1 = np.array([
            [0, 1, 2, 3],
            [1, 2, 3, 4],
            [2, 3, 4, 5],
            [3, 4, 5, 6]
        ])
        Y_TRAIN_CASE_1 = np.array([0, 1, 0, 1])
        STAY_PARAMS_1, LEAVE_PARAMS_1 = T.fit_gaussian_params(X_TRAIN_CASE_1,

        print("STAY PARAMETERS")
        for f_idx in range(len(STAY_PARAMS_1)):
            print(f"Feature : {f_idx}")
            print(f"Mean : {STAY_PARAMS_1[f_idx][0]}")
            print(f"STD. : {STAY_PARAMS_1[f_idx][1]}")
        print("")
        print("LEAVE PARAMETERS")
        for f_idx in range(len(LEAVE_PARAMS_1)):
            print(f"Feature : {f_idx}")
            print(f"Mean : {LEAVE_PARAMS_1[f_idx][0]}")
            print(f"STD. : {LEAVE_PARAMS_1[f_idx][1]}")

        check_fit_gaussian_params()
```

STAY PARAMETERS

Feature : 0
 Mean : 1.0
 STD. : 1.0
 Feature : 1
 Mean : 2.0
 STD. : 1.0
 Feature : 2
 Mean : 3.0
 STD. : 1.0
 Feature : 3
 Mean : 4.0
 STD. : 1.0

LEAVE PARAMETERS

Feature : 0
 Mean : 2.0
 STD. : 1.0
 Feature : 1
 Mean : 3.0
 STD. : 1.0
 Feature : 2
 Mean : 4.0
 STD. : 1.0
 Feature : 3
 Mean : 5.0
 STD. : 1.0

```
In [73]: y_pred = model.gaussian_predict(x_test)
```

```
In [74]: evaluate(y_test, y_pred)
```

```
Accuracy: 0.7581
Precision: 0.3750
Recall: 0.2308
F1 Score: 0.2857
FPR: 0.1020
```

```
Out[74]: (np.float64(0.7580645161290323),
          np.float64(0.375),
          np.float64(0.23076923076923078),
          np.float64(0.2857142857142857),
          np.float64(0.10204081632653061))
```

T13 : The random choice baseline is the accuracy if you make a random guess for each test sample. Give random guess (50% leaving, and 50% staying) to the test samples. Report the overall Accuracy. Then, report the Precision, Recall, and F score for attrition prediction using the random choice baseline.

```
In [75]: # T13: Random choice baseline
          np.random.seed(42)
          y_pred_random = np.random.choice([0, 1], size=len(y_test), p=[0.5, 0.5])
          print("Random Choice Baseline:")
          evaluate(y_test, y_pred_random)
```

```
Random Choice Baseline:
Accuracy: 0.3871
Precision: 0.1212
Recall: 0.3077
F1 Score: 0.1739
FPR: 0.5918
```

```
Out[75]: (np.float64(0.3870967741935484),
          np.float64(0.12121212121212122),
          np.float64(0.3076923076923077),
          np.float64(0.17391304347826086),
          np.float64(0.5918367346938775))
```

T14. The majority rule is the accuracy if you use the most frequent class from the training set as the classification decision. Report the overall Accuracy. Then, report the Precision, Recall, and F score for attrition prediction using the majority rule baseline.

```
In [76]: # T14: Majority rule baseline
          # The most frequent class in training set

          # Filter valid labels (0 and 1) to ignore any missing values encoded as -
          valid_train_mask = (y_train == 0) | (y_train == 1)
          y_train_clean = y_train[valid_train_mask]

          majority_class = int(np.round(np.mean(y_train_clean)))
```

```
y_pred_majority = np.full(len(y_test), majority_class)
print(f"Majority Rule Baseline (Class {majority_class}):")
evaluate(y_test, y_pred_majority)
```

Majority Rule Baseline (Class 0):

Accuracy: 0.7903

Precision: 0.0000

Recall: 0.0000

F1 Score: 0.0000

FPR: 0.0000

Out[76]: (np.float64(0.7903225806451613), 0, np.float64(0.0), 0, np.float64(0.0))

T15. Compare the two baselines with your Naive Bayes classifier.

T15 Answer: The Naive Bayes classifier should outperform both baselines. The random baseline achieves approximately 50% accuracy with low precision and recall. The majority baseline achieves accuracy equal to the proportion of the majority class but fails to detect the minority class (attrition). Our Naive Bayes classifier uses feature information to make informed predictions, resulting in better balanced performance across all metrics.

T16. Use the following threshold values

$t = np.arange(-5, 5, 0.05)$

find the best accuracy, and F score (and the corresponding thresholds)

```
In [77]: # T16: Find best threshold
t = np.arange(-5, 5, 0.05)
scores = model.predict_proba_scores(x_test, use_gaussian=False)

best_accuracy = 0
best_f1 = 0
best_threshold_acc = 0
best_threshold_f1 = 0

for threshold in t:
    y_pred_threshold = (scores > threshold).astype(int)
    accuracy, precision, recall, f1, fpr = evaluate(y_test, y_pred_thresh

    if accuracy > best_accuracy:
        best_accuracy = accuracy
        best_threshold_acc = threshold

    if f1 > best_f1:
        best_f1 = f1
        best_threshold_f1 = threshold

print(f"Best Accuracy: {best_accuracy:.4f} at threshold: {best_threshold_
print(f"Best F1 Score: {best_f1:.4f} at threshold: {best_threshold_f1:.4f
```

Best Accuracy: 0.8548 at threshold: -1.2000

Best F1 Score: 0.6087 at threshold: -1.2000

T17. Plot the RoC of your classifier.

```
In [78]: # Debug: Check unique values in y_test
print("Unique values in y_test:", np.unique(y_test))
print("y_test shape:", y_test.shape)
print("Number of NaN values:", np.sum(np.isnan(y_test)))
```

```
Unique values in y_test: [ 0.  1.  2.  3.  4.  5.  6.  7.  8.  9. 10. 11.
 13. 14. 15. 17.]
y_test shape: (294,)
Number of NaN values: 0
```

```
In [79]: # T17: Plot ROC curve
from sklearn.metrics import roc_curve, auc

# Filter for valid binary labels (0 or 1) and remove NaN values
valid_indices = ~np.isnan(y_test) & ((y_test == 0) | (y_test == 1))
y_test_clean = y_test[valid_indices].astype(int)
x_test_clean = x_test[valid_indices]

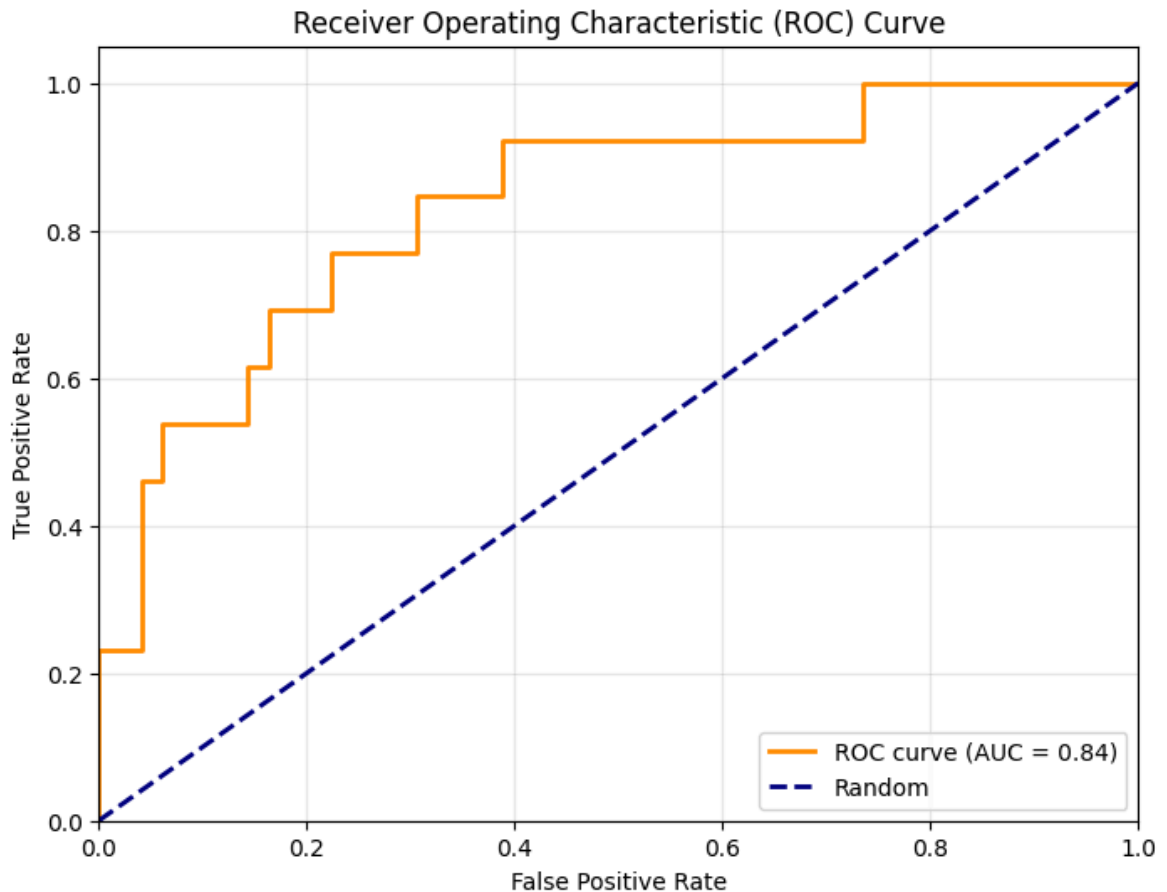
print(f"Using {len(y_test_clean)} samples out of {len(y_test)} total")
print(f"Unique values after filtering: {np.unique(y_test_clean)}")

# Get prediction scores
scores = model.predict_proba_scores(x_test_clean, use_gaussian=False)

# Calculate ROC curve
fpr_roc, tpr_roc, thresholds = roc_curve(y_test_clean, scores)
roc_auc = auc(fpr_roc, tpr_roc)

# Plot ROC curve
plt.figure(figsize=(8, 6))
plt.plot(fpr_roc, tpr_roc, color='darkorange', lw=2, label=f'ROC curve (AUC={roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--', label='Random')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.grid(True, alpha=0.3)
plt.show()
```

```
Using 62 samples out of 294 total
Unique values after filtering: [0 1]
```



T18. Change the number of discretization bins to 5. What happens to the RoC curve? Which discretization is better? The number of discretization bins can be considered as a hyperparameter, and must be chosen by comparing the final performance.

```
In [80]: # T18: Compare with 5 bins
model_5bins = SimpleBayesClassifier(n_pos=int(np.sum(y_train == 1)), n_neg=int(np.sum(y_train == 0)), n_bins=5)
model_5bins.fit_params(x_train, y_train)

# Filter for valid binary labels (0 or 1) and remove NaN values
valid_indices = ~np.isnan(y_test) & ((y_test == 0) | (y_test == 1))
y_test_clean = y_test[valid_indices].astype(int)
x_test_clean = x_test[valid_indices]

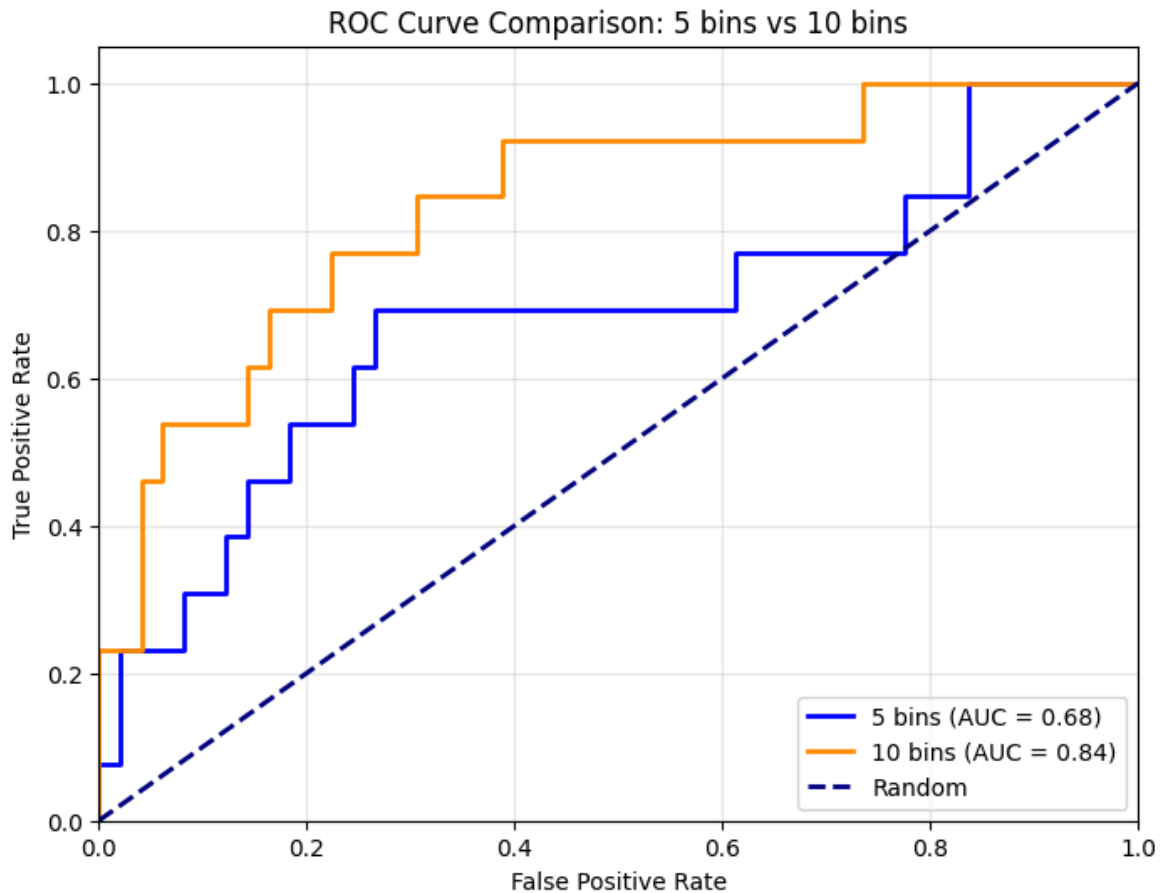
# Get scores for 5 bins model
scores_5bins = model_5bins.predict_proba_scores(x_test_clean, use_gaussian=False)
fpr_5bins, tpr_5bins, _ = roc_curve(y_test_clean, scores_5bins)
roc_auc_5bins = auc(fpr_5bins, tpr_5bins)

# Get scores for 10 bins model
scores_10bins = model.predict_proba_scores(x_test_clean, use_gaussian=False)
fpr_10bins, tpr_10bins, _ = roc_curve(y_test_clean, scores_10bins)
roc_auc_10bins = auc(fpr_10bins, tpr_10bins)

# Plot comparison
plt.figure(figsize=(8, 6))
plt.plot(fpr_5bins, tpr_5bins, color='blue', lw=2, label=f'5 bins (AUC = {roc_auc_5bins:.2f})')
plt.plot(fpr_10bins, tpr_10bins, color='darkorange', lw=2, label=f'10 bin (AUC = {roc_auc_10bins:.2f})')
```

```
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--', label='Rando')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison: 5 bins vs 10 bins')
plt.legend(loc="lower right")
plt.grid(True, alpha=0.3)
plt.show()

print(f"\n5 bins AUC: {roc_auc_5bins:.4f}")
print(f"\n10 bins AUC: {roc_auc_10bins:.4f}")
print(f"\nThe discretization with {'5' if roc_auc_5bins > roc_auc_10bins
```



5 bins AUC: 0.6813
 10 bins AUC: 0.8352

The discretization with 10 bins performs better based on AUC.